LESSON 4.5

Determine the Whole Given the Part and Percent

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can determine the whole when I know a part and the percent that part represents.

WHY THIS MATTERS Every percent problem has three pieces: the part, the whole, and the percent. So far the whole has always been handed to you. Today it is the missing piece — a sale price without the original price, a car's current value without its sticker price. Your job is to work backwards to the whole. For example, Akela paid $42 for a sweater on sale, and that was 60% of the original price. What was the original price?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Part *Spanish: Parte*	The amount you already know — the piece of the whole that the percent describes.	$42 is the sale price
Whole *Spanish: Total*	The full amount the percent is measured against. The whole is always 100%.	the original price
Double number line *Spanish: Recta numérica doble*	Two lines lined up — dollars on one, percents on the other — so matching marks show the same amount two ways.	$0, $7, $70 · 0%, 10%, 100%
Equation *Spanish: Ecuación*	A math sentence with an equal sign. Here it says percent × whole = part, with the whole unknown.	0.7v = 26,600 — the percent times the unknown whole equals the part
Percent *Spanish: Porcentaje*	A ratio comparing a number to 100. To compute with it, write it in decimal form: 60% = 0.6.	60% = 0.6 · 70% = 0.7
Inverse operation *Spanish: Operación inversa*	The operation that undoes another operation.	Division undoes multiplication.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

24 stickers are 40% of a pack. How many stickers are in the pack?

1. 40% ÷ 4 = 10%, so 24 ÷ 4 = 6.
2. 10% × 10 = 100%, so 6 × 10 = 60.
3. Check: 0.40 × 60 = 24.

Answer: 60 stickers

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Sheng's car is now worth $26,600, which is 70% of what his parents paid for it. Find the original price.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The part is _____ and it matches _____ %.
- Ten percent of the whole is _____ .
- The whole is _____ because _____ .
- I checked my answer by finding _____ % of it.

Word bank part | whole | percent | double number line | equivalent ratio | scale | divide | multiply | ten percent | one hundred percent | table | check

WATCH OUT!

A common mistake

A common mistake when finding the whole is multiplying by the percent instead of dividing. For "$42 is 60% of the original price," computing 0.6 × 42 = $25.20 makes the original price SMALLER than the sale price

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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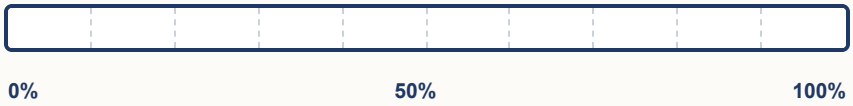
1 **Circle the letter of the best answer. Show how you know.**

$20 is 50% of what number?

(A) $10
(B) $25
(C) $40
(D) $70

STEPS TO TRY

1. 50% means half of the whole.
2. If half the whole is 20, what is the whole?
3. 20 ÷ 0.5 = ?



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

2 **Circle the letter of the best answer. Show how you know.**

Which equation says "30 is 25% of some number n"?

(A) 0.25n = 30
(B) n = 0.25 × 30
(C) 25n = 30
(D) n = 30 + 25

STEPS TO TRY

1. Which number is the part, and which is the percent?
2. Percent × whole = part.
3. Write 25% as 0.25 and let n be the whole.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

$42 is 60% of the original price. Should the original price be more or less than $42?

(A) More, because 60% is less than the whole

(B) Less, because 60 is a big number

(C) Exactly $42

(D) You cannot tell

STEPS TO TRY

1. What percent stands for the whole?
2. Is 60% more or less than 100%?
3. A part is smaller than its whole.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

75 minutes is 15% of the school day. How long is the school day?

(A) 11.25 minutes

(B) 90 minutes

(C) 500 minutes

(D) 1,125 minutes

STEPS TO TRY

1. 15% is three groups of 5%. Split 75 minutes into 3 equal groups to find 5%.
2. 75 ÷ 3 = 25, so 5% of the day is 25 minutes. How many 5% groups make 100%?
3. 100 ÷ 5 = 20 groups. Then 25 × 20 = 500 minutes. Check: 10% of 500 is 50 and 5% is 25; 50 + 25 = 75.



0% 50% 100%

WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

EXPLAIN YOUR THINKING

How can you tell, before doing any arithmetic, whether the answer should be bigger or smaller than the number you were given?

SENTENCE STARTER If the percent is less than 100%, the given number is a ___, so the whole must be ___ than it.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help